Adjusting published removal anomalies

Let $Q_j > 0$ denote annual methane removal by OH, m_j the atmospheric burden, and $k_j = Q_j/m_j$ the effective loss frequency. The reference year is 2019, denoted by subscript 0. He's OH forcing is

$$F_{\mathrm{OH},j} = -m_0(k_j - k_0) = Q_0 - Q_j m_0/m_j. \tag{1}$$

Qu (2022), Table 1, supplies both Q and m. We use their ratio consistently, rather than mixing it with the separately reported lifetime column. Its other removal combines chemical and soil sinks, which cannot be separated into He's two terms from the table.

Ciais (2026), Figure S3, plots signed sink anomalies $S = -Q$ relative to 2010. Subtracting each curve's 2019 value gives

$$s_j = S_j - S_0 = Q_0 - Q_j. \tag{2}$$

The 2010 reference cancels. Native Q_0 and annual burdens are not tabulated. For a conditional comparison we prescribe the NOAA surface-mean burden proxy and an OH-only lifetime $\tau_{\mathrm{OH},0}$:

$$Q_0 = \frac{m_0}{\tau_{\mathrm{OH},0}}, \qquad F_{\mathrm{OH},j} = \frac{m_0}{m_j}\left(s_j + \frac{m_j - m_0}{\tau_{\mathrm{OH},0}}\right). \tag{3}$$

This sensitivity calculation does not recover Ciais's native forcing. We use 11.1 yr as a benchmark and 9–12 yr as a sensitivity range; 11.1 yr is He's reported mean OH lifetime, not a known Ciais 2019 value. Coordinate-reading precision is separate from inversion spread. The annual circles are read at their published year labels.

For a uniform fractional OH reduction r, with unchanged temperature and baseline weighting, the prescribed scenario gives

$$k_j = (1 - r)k_0, \qquad F_{\mathrm{OH},j} = r\, m_0/\tau_{\mathrm{OH},0}. \tag{4}$$

Laughner's 2–4% range motivates r; it is a temporal sensitivity envelope, not a posterior confidence interval. The reduction is applied during 2020 only. Returning the imposed forcing to zero in 2021–2022 does not remove the burden accumulated in 2020. This calculation does not reproduce Laughner's hemispheric trajectories or their 2021 extrapolations. All flux terms have units Tg CH_4 yr^{-1} and burdens Tg CH_4.